

```
jismiya@qbs-generic-lin-2204:~/LINUX/hello$ sudo dmesg -T
[Thu Jun 11 12:11:46 2026] hello          Module Loaded
jismiya@qbs-generic-lin-2204:~/LINUX/hello$ sudo dmesg --buffer -size
dmesg: invalid buffer size argument: '-size'
jismiya@qbs-generic-lin-2204:~/LINUX/hello$ cat /proc/sys/kernel/printk
4      4      1      7
jismiya@qbs-generic-lin-2204:~/LINUX/hello$ uname -r
6.8.0-124-generic
jismiya@qbs-generic-lin-2204:~/LINUX/hello$ uname -a
Linux qbs-generic-lin-2204 6.8.0-124-generic #124~22.04.1-Ubuntu SMP PREEMPT_DYNAMIC Tue May 26 21:05:19 UTC x86_64 x86_64 x86_64 GNU/Linux
jismiya@qbs-generic-lin-2204:~/LINUX/hello$ |
```

```

jismiya@qbs-generic-lin-2204:~/LINUX/BUG_ON$ ls
bug_on.c bug_on.ko bug_on.mod bug_on.mod.c bug_on.mod.o bug_on.o Makefile modules.order Module.symvers
jismiya@qbs-generic-lin-2204:~/LINUX/BUG_ON$ cat bug_on.c
#include <linux/module.h>
#include <linux/kernel.h>

static int __init my_init(void)
{
    int value = 10;

    pr_info("Before BUG_ON()\n");

    BUG_ON(value == 10);

    pr_info("After BUG_ON()\n");

    return 0;
}

static void __exit my_exit(void)
{
    pr_info("Module Removed\n");
}

module_init(my_init);
module_exit(my_exit);

MODULE_LICENSE("GPL");
jismiya@qbs-generic-lin-2204:~/LINUX/BUG_ON$ sudo insmod bug_on.ko
Segmentation fault (core dumped)
jismiya@qbs-generic-lin-2204:~/LINUX/BUG_ON$ sudo dmesg |tail
[113242.150934] Code: 90 90 90 90 90 90 90 90 90 90 90 90 90 90 0f 1f 44 00 00 55 48 c7 c7 54 b0 6a c0 48 89 e5 e8 9b 7e 70
[113242.150938] RSP: 0018:ffffd2c180ddb830 EFLAGS: 00010246
[113242.150939] RAX: 000000000000000b RBX: ffffffff06ad010 RCX: 0000000000000000
[113242.150939] RDX: 0000000000000000 RSI: 0000000000000000 RDI: ffffffff06ab063
[113242.150940] RBP: fffffd2c180ddb830 R08: 0000000000000000 R09: 0000000000000000
[113242.150941] R10: 0000000000000000 R11: 0000000000000000 R12: 0000000000000000
[113242.150941] R13: 0000000000000000 R14: fffffd2c180ddb840 R15: ffffffff06a9040
[113242.150942] FS: 00007ee137f3fc40(0000) GS:ffff89f5caa80000(0000) knlGS:0000000000000000
[113242.150943] CS: 0010 DS: 0000 ES: 0000 CR0: 0000000080050033
[113242.150944] CR2: ffffffff06b6fffb CR3: 000000002704a000 CR4: 00000000000006f0
jismiya@qbs-generic-lin-2204:~/LINUX/BUG_ON$ sudo rmmod bug_on
rmmod: ERROR: Module bug_on is in use
jismiya@qbs-generic-lin-2204:~/LINUX/BUG_ON$ sudo dmesg |tail
[113242.150934] Code: 90 90 90 90 90 90 90 90 90 90 90 90 90 90 0f 1f 44 00 00 55 48 c7 c7 54 b0 6a c0 48 89 e5 e8 9b 7e 70
[113242.150938] RSP: 0018:ffffd2c180ddb830 EFLAGS: 00010246
[113242.150939] RAX: 000000000000000b RBX: ffffffff06ad010 RCX: 0000000000000000
[113242.150939] RDX: 0000000000000000 RSI: 0000000000000000 RDI: ffffffff06ab063
[113242.150940] RBP: fffffd2c180ddb830 R08: 0000000000000000 R09: 0000000000000000
[113242.150941] R10: 0000000000000000 R11: 0000000000000000 R12: 0000000000000000
[113242.150941] R13: 0000000000000000 R14: fffffd2c180ddb840 R15: ffffffff06a9040
[113242.150942] FS: 00007ee137f3fc40(0000) GS:ffff89f5caa80000(0000) knlGS:0000000000000000
[113242.150943] CS: 0010 DS: 0000 ES: 0000 CR0: 0000000080050033
[113242.150944] CR2: ffffffff06b6fffb CR3: 000000002704a000 CR4: 00000000000006f0
jismiya@qbs-generic-lin-2204:~/LINUX/BUG_ON$ sudo dmesg -C
jismiya@qbs-generic-lin-2204:~/LINUX/BUG_ON$ sudo dmesg
jismiya@qbs-generic-lin-2204:~/LINUX/BUG_ON$ |

```

```

jismiya@qbs-generic-lin-2204:~/LINUX/CURRENT_PID$ cat current_pid.c
#include <linux/module.h>
#include <linux/kernel.h>
#include <linux/sched.h>

static int __init my_init(void)
{
    pr_info("Current PID = %d\n", current->pid);

    return 0;
}

static void __exit my_exit(void)
{
    pr_info("Module Removed\n");
}

module_init(my_init);
module_exit(my_exit);

MODULE_LICENSE("GPL");
jismiya@qbs-generic-lin-2204:~/LINUX/CURRENT_PID$ sudo insmod current_pid.ko
jismiya@qbs-generic-lin-2204:~/LINUX/CURRENT_PID$ sudo dmesg |tail
[112974.886996] Current PID = 40397
jismiya@qbs-generic-lin-2204:~/LINUX/CURRENT_PID$ sudo rmmod current_pid
jismiya@qbs-generic-lin-2204:~/LINUX/CURRENT_PID$ sudo dmesg |tail
[112974.886996] Current PID = 40397
[113008.210840] Module Removed
jismiya@qbs-generic-lin-2204:~/LINUX/CURRENT_PID$ sudo dmesg -C
jismiya@qbs-generic-lin-2204:~/LINUX/CURRENT_PID$ sudo dmesg
jismiya@qbs-generic-lin-2204:~/LINUX/CURRENT_PID$ |

```

```

jismiya@qbs-generic-lin-2204:~/LINUX/dump_stack$ cat stack.c
#include <linux/module.h>
#include <linux/kernel.h>

static int __init my_init(void)
{
    pr_info("Before dump_stack()\n");

    dump_stack();

    pr_info("After dump_stack()\n");

    return 0;
}

static void __exit my_exit(void)
{
    pr_info("Module Removed\n");
}

module_init(my_init);
module_exit(my_exit);

MODULE_LICENSE("GPL");
jismiya@qbs-generic-lin-2204:~/LINUX/dump_stack$ sudo insmod stack.ko
jismiya@qbs-generic-lin-2204:~/LINUX/dump_stack$ sudo dmesg |tail
[112807.435976] RIP: 0033:0x77c5e231e90d
[112807.435990] Code: 5b 41 5c c3 66 0f 1f 84 00 00 00 00 00 f3 0f 1e fa 48 89 f8 48 89 f7 48 89 d6 48
[112807.435991] RSP: 002b:00007ffdb6e86528 EFLAGS: 00000246 ORIG_RAX: 0000000000000139
[112807.435994] RAX: ffffffffefda RBX: 000062f88c2af780 RCX: 000077c5e231e90d
[112807.435995] RDX: 0000000000000000 RSI: 000062f87476bcd2 RDI: 0000000000000003
[112807.435996] RBP: 0000000000000000 R08: 0000000000000000 R09: 0000000000000000
[112807.435999] R10: 0000000000000003 R11: 0000000000000246 R12: 000062f87476bcd2
[112807.436000] R13: 000062f88c2b3770 R14: 000062f87476a888 R15: 000062f88c2af890
[112807.436001] </TASK>
[112807.436002] After dump_stack()
jismiya@qbs-generic-lin-2204:~/LINUX/dump_stack$ sudo rmmod stack
jismiya@qbs-generic-lin-2204:~/LINUX/dump_stack$ sudo dmesg |tail
[112807.435990] Code: 5b 41 5c c3 66 0f 1f 84 00 00 00 00 00 f3 0f 1e fa 48 89 f8 48 89 f7 48 89 d6 48
[112807.435991] RSP: 002b:00007ffdb6e86528 EFLAGS: 00000246 ORIG_RAX: 0000000000000139
[112807.435994] RAX: ffffffffefda RBX: 000062f88c2af780 RCX: 000077c5e231e90d
[112807.435995] RDX: 0000000000000000 RSI: 000062f87476bcd2 RDI: 0000000000000003
[112807.435996] RBP: 0000000000000000 R08: 0000000000000000 R09: 0000000000000000
[112807.435999] R10: 0000000000000003 R11: 0000000000000246 R12: 000062f87476bcd2
[112807.436000] R13: 000062f88c2b3770 R14: 000062f87476a888 R15: 000062f88c2af890
[112807.436001] </TASK>
[112807.436002] After dump_stack()
[112828.296177] Module Removed
jismiya@qbs-generic-lin-2204:~/LINUX/dump_stack$ sudo dmesg -C
jismiya@qbs-generic-lin-2204:~/LINUX/dump_stack$ sudo dmesg
jismiya@qbs-generic-lin-2204:~/LINUX/dump_stack$ |

```

```
jismiya@qbs-generic-lin-2204:~/LINUX/Dynamic_debug$ ls
debug.c debug.ko debug.mod debug.mod.c debug.mod.o debug.o Makefile modules.order Module.symvers
jismiya@qbs-generic-lin-2204:~/LINUX/Dynamic_debug$ cat debug.c
#define DEBUG
#include <linux/module.h>
#include <linux/kernel.h>

static int __init my_init(void)
{
    pr_debug("Debug Message\n");
    pr_info("Info Message\n");

    return 0;
}

static void __exit my_exit(void)
{
    pr_info("Module Removed\n");
}

module_init(my_init);
module_exit(my_exit);

MODULE_LICENSE("GPL");
jismiya@qbs-generic-lin-2204:~/LINUX/Dynamic_debug$ sudo insmod debug.ko
jismiya@qbs-generic-lin-2204:~/LINUX/Dynamic_debug$ sudo dmesg | tail
[113456.043424] Debug Message
[113456.043451] Info Message
jismiya@qbs-generic-lin-2204:~/LINUX/Dynamic_debug$ sudo rmmod debug
jismiya@qbs-generic-lin-2204:~/LINUX/Dynamic_debug$ sudo dmesg |tail
[113456.043424] Debug Message
[113456.043451] Info Message
[113494.242311] Module Removed
jismiya@qbs-generic-lin-2204:~/LINUX/Dynamic_debug$ sudo dmesg -C
jismiya@qbs-generic-lin-2204:~/LINUX/Dynamic_debug$ sudo dmesg
jismiya@qbs-generic-lin-2204:~/LINUX/Dynamic_debug$ |
```

```
jismiya@qbs-generic-lin-2204 x + v
jismiya@qbs-generic-lin-2204:~/LINUX/gcc-alias$ cat gcc_alias.c
#include <linux/module.h>
#include <linux/kernel.h>

void hello(void)
{
    printk(KERN_INFO "Hello Function\n");
}

void test(void)
__attribute__((alias("hello")));

static int hello_init(void)
{
    test();
    return 0;
}

static void hello_exit(void)
{
    printk(KERN_INFO "Module Removed\n");
}

module_init(hello_init);
module_exit(hello_exit);

MODULE_LICENSE("GPL");
jismiya@qbs-generic-lin-2204:~/LINUX/gcc-alias$ sudo dmesg -C
jismiya@qbs-generic-lin-2204:~/LINUX/gcc-alias$ sudo insmod gcc_alias.ko
jismiya@qbs-generic-lin-2204:~/LINUX/gcc-alias$ sudo dmesg | tail
[90576.723187] Hello Function
jismiya@qbs-generic-lin-2204:~/LINUX/gcc-alias$ sudo rmmod gcc_alias
jismiya@qbs-generic-lin-2204:~/LINUX/gcc-alias$ sudo dmesg -C
jismiya@qbs-generic-lin-2204:~/LINUX/gcc-alias$ |
```

Activate Windows
Go to Settings to activate Windows.



12:24
11-06-2026

jismiya@qbs-generic-lin-2204

```
jismiya@qbs-generic-lin-2204:~/LINUX/hello$ sudo insmod helo.ko
[sudo] password for jismiya:
jismiya@qbs-generic-lin-2204:~/LINUX/hello$ sudo dmesg | tail
[89798.355557] hello    Module Loaded
jismiya@qbs-generic-lin-2204:~/LINUX/hello$ lsmod helo.ko
Usage: lsmod
jismiya@qbs-generic-lin-2204:~/LINUX/hello$ sudo rmmod helo
jismiya@qbs-generic-lin-2204:~/LINUX/hello$ sudo dmesg | tail
[89798.355557] hello    Module Loaded
[89832.236527] hello Module Removed
jismiya@qbs-generic-lin-2204:~/LINUX/hello$ sudo dmesg -c
[89798.355557] hello    Module Loaded
[89832.236527] hello Module Removed
jismiya@qbs-generic-lin-2204:~/LINUX/hello$ sudo dmesg -C
jismiya@qbs-generic-lin-2204:~/LINUX/hello$ sudo dmesg
jismiya@qbs-generic-lin-2204:~/LINUX/hello$ sudo insmod helo.ko
jismiya@qbs-generic-lin-2204:~/LINUX/hello$ modinfo helo.ko
filename:        /home/jismiya/LINUX/hello/helo.ko
description:     My first kernel Module
author:          jismiya
license:         GPL
srcversion:      F7BE6ACEC2A390A48DEFE55
depends:
retpoline:      Y
name:           helo
vermagic:       6.8.0-124-generic SMP preempt mod_unload modversions
jismiya@qbs-generic-lin-2204:~/LINUX/hello$ |
```

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 1



ENG
IN



12:12
11-06-2026

```
jismiya@qbs-generic-lin-2204:~/LINUX/KER_RING_BUFF$ cat buffer.c
#include <linux/module.h>
```

```
static int __init my_init(void)
{
    printk(KERN_INFO "Ring Buffer Demo\n");
    return 0;
}
```

```
static void __exit my_exit(void)
{
    printk(KERN_INFO "Module Removed\n");
}
```

```
module_init(my_init);
module_exit(my_exit);
MODULE_LICENSE("GPL");
```

/// In this ring buffer if we type the command dmesg |tail it will print eve//rything that you have done in all diretieris because everything getting stor//ed in kernel ring buffer

```
jismiya@qbs-generic-lin-2204:~/LINUX/KER_RING_BUFF$ sudo insmod buffer.ko
```

```
jismiya@qbs-generic-lin-2204:~/LINUX/KER_RING_BUFF$ sudo dmesg |tail
```

```
[113871.138147] Ring Buffer Demo
```

```
jismiya@qbs-generic-lin-2204:~/LINUX/KER_RING_BUFF$ sudo rmmod buffer
```

```
jismiya@qbs-generic-lin-2204:~/LINUX/KER_RING_BUFF$ sudo dmesg |tail
```

```
[113871.138147] Ring Buffer Demo
```

```
[113891.071453] Module Removed
```

```
jismiya@qbs-generic-lin-2204:~/LINUX/KER_RING_BUFF$ sudo dmesg -C
```

```
jismiya@qbs-generic-lin-2204:~/LINUX/KER_RING_BUFF$ sudo dmesg
```

```
jismiya@qbs-generic-lin-2204:~/LINUX/KER_RING_BUFF$ |
```



```

jismiya@qbs-generic-lin-2204:~/LINUX/METADATA$ ls
hello.c hello.ko hello.mod hello.mod.c hello.mod.o hello.o Makefile modules.order Module.symvers
jismiya@qbs-generic-lin-2204:~/LINUX/METADATA$ cat hello.c
#include <linux/module.h>
#include <linux/kernel.h>

MODULE_LICENSE("GPL");
MODULE_AUTHOR("Olga Evelyn");
MODULE_DESCRIPTION("Learning Linux Kernel Modules");
MODULE_INFO(batch, "LinuxKT");
MODULE_INFO(company, "Emertxe");
MODULE_VERSION("1.0");

static int __init hello_init(void)
{
    printk(KERN_INFO "Metadata Module Loaded\n");
    return 0;
}

static void __exit hello_exit(void)
{
    printk(KERN_INFO "Metadata Module Removed\n");
}

module_init(hello_init);
module_exit(hello_exit);

//Metadata means all module eraleated informations a re stored in .modinfo$//ace if wwe want to see all things just type
//modinfo hello.ko
//
//
//Later i added the term modinfo below author and description
jismiya@qbs-generic-lin-2204:~/LINUX/METADATA$ modinfo hello.ko
filename:       /home/jismiya/LINUX/METADATA/hello.ko
version:       1.0
company:       Emertxe
batch:        LinuxKT
description:   Learning Linux Kernel Modules
author:       Olga Evelyn
license:      GPL
srcversion:    A30DDAC8C9DE134E732F6A4
depends:
retpoline:    Y
name:         hello
vermagic:     6.8.0-124-generic SMP preempt mod_unload modversions
jismiya@qbs-generic-lin-2204:~/LINUX/METADATA$ |

```

```

jismiya@qbs-generic-lin-2204:~/Linux/Export_Symbol$ ls
Makefile  mod_A.ko  mod_A.mod.c  mod_A.o  mod_B.ko  mod_B.mod.c  mod_B.o  Module.symvers
mod_A.c   mod_A.mod  mod_A.mod.o  mod_B.c  mod_B.mod  mod_B.mod.o  modules.order
jismiya@qbs-generic-lin-2204:~/Linux/Export_Symbol$ cat mod_A.c
// module_a.c
#include <linux/module.h>

int add(int a, int b)
{
    return a + b;
}

EXPORT_SYMBOL(add);

static int __init my_init(void)
{
    printk(KERN_INFO "Module A Loaded\n");
    return 0;
}

static void __exit my_exit(void)
{
    printk(KERN_INFO "Module A Removed\n");
}

module_init(my_init);
module_exit(my_exit);

MODULE_LICENSE("GPL");
jismiya@qbs-generic-lin-2204:~/Linux/Export_Symbol$ cat mod_B.c
// module_b.c
#include <linux/module.h>

extern int add(int, int);

static int __init my_init(void)
{
    printk(KERN_INFO "Result=%d\n", add(10,20));
    return 0;
}

static void __exit my_exit(void)
{
}

module_init(my_init);
module_exit(my_exit);

MODULE_LICENSE("GPL");
jismiya@qbs-generic-lin-2204:~/Linux/Export_Symbol$ sudo insmod mod_A.ko
jismiya@qbs-generic-lin-2204:~/Linux/Export_Symbol$ sudo dmesg | tail
[90722.141837] Module A Loaded
jismiya@qbs-generic-lin-2204:~/Linux/Export_Symbol$ sudo insmod mod_B.ko
jismiya@qbs-generic-lin-2204:~/Linux/Export_Symbol$ sudo dmesg | tail
[90722.141837] Module A Loaded
[90747.080094] Result=30
jismiya@qbs-generic-lin-2204:~/Linux/Export_Symbol$ sudo rmmod mod_A
rmmod: ERROR: Module mod_A is in use by: mod_B
jismiya@qbs-generic-lin-2204:~/Linux/Export_Symbol$ sudo rmmod mod_B
jismiya@qbs-generic-lin-2204:~/Linux/Export_Symbol$ sudo rmmod mod_A
jismiya@qbs-generic-lin-2204:~/Linux/Export_Symbol$ sudo dmesg | tail
[90722.141837] Module A Loaded
[90747.080094] Result=30
[90777.829787] Module A Removed
jismiya@qbs-generic-lin-2204:~/Linux/Export_Symbol$ |

```

```
jismiya@qbs-generic-lin-2204:~/LINUX/OOPS$ cat oops.c
```

```
#include <linux/module.h>
```

```
#include <linux/kernel.h>
```

```
static int __init my_init(void)
```

```
{
```

```
    int *ptr = NULL;
```

```
    pr_info("Before Oops\n");
```

```
    *ptr = 100;
```

```
    pr_info("After Oops\n");
```

```
    return 0;
```

```
}
```

```
static void __exit my_exit(void)
```

```
{
```

```
    pr_info("Module Removed\n");
```

```
}
```

```
module_init(my_init);
```

```
module_exit(my_exit);
```

```
MODULE_LICENSE("GPL");
```

```
jismiya@qbs-generic-lin-2204:~/LINUX/OOPS$ sudo insmod oops.ko
```

```
Killed
```

```
jismiya@qbs-generic-lin-2204:~/LINUX/OOPS$ sudo dmesg |tail
```

```
[111863.403162] Code: 90 90 90 90 90 90 90 90 90 90 90 90 90 0f 1f 44 00 00 55 48 c7 c7 54  
7e 70 c3 31 c0 5d 31 ff c3
```

```
[111863.403163] RSP: 0018:ffffd2c180ddb830 EFLAGS: 00010246
```

```
[111863.403164] RAX: 0000000000000000b RBX: ffffffffcc06ad010 RCX: 0000000000000000
```

```
[111863.403164] RDX: 0000000000000000 RSI: 0000000000000000 RDI: ffffffffcc06ab063
```

```
[111863.403165] RBP: fffffd2c180ddb830 R08: 0000000000000000 R09: 0000000000000000
```

```
[111863.403166] R10: 0000000000000000 R11: 0000000000000000 R12: 0000000000000000
```

```
[111863.403166] R13: 0000000000000000 R14: fffffd2c180ddb840 R15: ffffffffcc06a9040
```

```
[111863.403167] FS: 00007902e54b8c40(0000) GS:ffff89f5cab00000(0000) knlGS:0000000000000000
```

```
[111863.403168] CS: 0010 DS: 0000 ES: 0000 CR0: 0000000080050033
```

```
[111863.403169] CR2: 0000000000000000 CR3: 0000000052fea000 CR4: 000000000000006f0
```

```
jismiya@qbs-generic-lin-2204:~/LINUX/OOPS$ sudo rmmod oop
```

```
rmmod: ERROR: Module oop is not currently loaded
```

```
jismiya@qbs-generic-lin-2204:~/LINUX/OOPS$ sudo dmesg -C
```

```
jismiya@qbs-generic-lin-2204:~/LINUX/OOPS$ sudo dmesg -C
```

```

jismiya@qbs-generic-lin-2204:~/LINUX/print_hex$ ls
hexdump.c hexdump.ko hexdump.mod hexdump.mod.c hexdump.mod.o hexdump.o Makefile modules.order Module.symvers
jismiya@qbs-generic-lin-2204:~/LINUX/print_hex$ cat hexdump.c
#include <linux/module.h>
#include <linux/kernel.h>

static unsigned char data[] = {
    0x11,
    0x22,
    0x33,
    0x44,
    0x55,
    0x66,
    0x77,
    0x88
};

static int __init my_init(void)
{
    print_hex_dump(KERN_INFO,
                    "DATA: ",
                    DUMP_PREFIX_OFFSET,
                    16,
                    1,
                    data,
                    sizeof(data),
                    true);

    return 0;
}

static void __exit my_exit(void)
{
    pr_info("Module Removed\n");
}

module_init(my_init);
module_exit(my_exit);

MODULE_LICENSE("GPL");
jismiya@qbs-generic-lin-2204:~/LINUX/print_hex$ sudo insmod hexdump.ko
jismiya@qbs-generic-lin-2204:~/LINUX/print_hex$ sudo dmesg | tail
[91367.787511] DATA: 00000000: 11 22 33 44 55 66 77 88                ."3DUfw.
jismiya@qbs-generic-lin-2204:~/LINUX/print_hex$ sudo rmmod hexdump
jismiya@qbs-generic-lin-2204:~/LINUX/print_hex$ sudo dmesg | tail
[91367.787511] DATA: 00000000: 11 22 33 44 55 66 77 88                ."3DUfw.
[91395.721483] Module Removed
jismiya@qbs-generic-lin-2204:~/LINUX/print_hex$ sudo dmesg -C
jismiya@qbs-generic-lin-2204:~/LINUX/print_hex$ sudo dmesg -C
jismiya@qbs-generic-lin-2204:~/LINUX/print_hex$ |

```

```

jismiya@qbs-generic-lin-2204:~/LINUX/1$
jismiya@qbs-generic-lin-2204:~/LINUX/1$ cat param.c
#include <linux/module.h>
#include <linux/kernel.h>
#include <linux/moduleparam.h>

int num = 0;
char *name = "Guest";

//module_param(num, int, 0644);
module_param(name, charp, 0644);

static int param_init(void)
{
    printk(KERN_INFO "Module Loaded\n");
    //    printk(KERN_INFO "num = %d\n", num);
    printk(KERN_INFO "name = %s\n", name);
    return 0;
}

static void param_exit(void)
{
    printk(KERN_INFO "Module Removed\n");
}

module_init(param_init);
module_exit(param_exit);

MODULE_LICENSE("GPL");
MODULE_AUTHOR("Jismiya");
MODULE_DESCRIPTION("Module Parameter Example");
jismiya@qbs-generic-lin-2204:~/LINUX/1$ sudo insmod param.ko value=50
jismiya@qbs-generic-lin-2204:~/LINUX/1$ sudo dmesg |tail
[111472.138675] Module Removed
[111482.924225] param: unknown parameter 'value' ignored
[111482.924270] Module Loaded
[111482.924270] num = 0
[111482.924270] name = Guest
[111565.267605] Module Removed
[111586.387315] param: unknown parameter 'value' ignored
[111586.387358] Module Loaded
[111586.387365] num = 0
[111586.387365] name = Guest
jismiya@qbs-generic-lin-2204:~/LINUX/1$ sudo rmmod param
jismiya@qbs-generic-lin-2204:~/LINUX/1$ sudo dmesg -C
jismiya@qbs-generic-lin-2204:~/LINUX/1$ sudo dmesg |tail

```

jismiya@qbs-generic-lin-2204

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jismiya@qbs-generic-lin-2204:~/LINUX/LOG_LEVELS\$ cat loglevel.c

#include <linux/module.h>

#include <linux/kernel.h>

static int __init my_init(void)

{

printk(KERN_EMERG "EMERG Message\n");

printk(KERN_ALERT "ALERT Message\n");

printk(KERN_CRIT "CRITICAL Message\n");

printk(KERN_ERR "ERROR Message\n");

printk(KERN_WARNING "WARNING Message\n");

printk(KERN_NOTICE "NOTICE Message\n");

printk(KERN_INFO "INFO Message\n");

printk(KERN_DEBUG "DEBUG Message\n");

return 0;

}

static void __exit my_exit(void)

{

printk(KERN_INFO "Module Removed\n");

}

module_init(my_init);

module_exit(my_exit);

MODULE_LICENSE("GPL");

jismiya@qbs-generic-lin-2204:~/LINUX/LOG_LEVELS\$ sudo insmod loglevel.ko

jismiya@qbs-generic-lin-2204:~/LINUX/LOG_LEVELS\$ sudo dmesg | tail

[90380.910658] DEBUG Message

[90392.298478] Module Removed

[90399.456722] EMERG Message

[90399.456728] ALERT Message

[90399.456729] CRITICAL Message

[90399.456729] ERROR Message

[90399.456730] WARNING Message

[90399.456730] NOTICE Message

[90399.456730] INFO Message

[90399.456731] DEBUG Message

jismiya@qbs-generic-lin-2204:~/LINUX/LOG_LEVELS\$ |

Activate Windows

Go to Settings to activate Windows.

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```

jismiya@qbs-generic-lin-2204:~/LINUX/pr_info$ ls
Makefile modules.order Module.symvers pr.c pr.ko pr.mod pr.mod.c pr.mod.o pr.o
jismiya@qbs-generic-lin-2204:~/LINUX/pr_info$ cat pr.c
#include <linux/module.h>
#include <linux/kernel.h>

static int __init my_init(void)
{
    pr_info("Info Message\n");

    pr_warn("Warning Message\n");

    pr_err("Error Message\n");

    return 0;
}

static void __exit my_exit(void)
{
    pr_info("Module Removed\n");
}

module_init(my_init);
module_exit(my_exit);

MODULE_LICENSE("GPL");
jismiya@qbs-generic-lin-2204:~/LINUX/pr_info$ sudo insmod pr.ko
jismiya@qbs-generic-lin-2204:~/LINUX/pr_info$ sudo dmesg |tail
[91011.135747] Info Message
[91011.135750] Warning Message
[91011.135750] Error Message
jismiya@qbs-generic-lin-2204:~/LINUX/pr_info$ sudo rmmod pr
jismiya@qbs-generic-lin-2204:~/LINUX/pr_info$ sudo dmesg |tail
[91011.135747] Info Message
[91011.135750] Warning Message
[91011.135750] Error Message
[91026.457964] Module Removed
jismiya@qbs-generic-lin-2204:~/LINUX/pr_info$ sudo dmesg -C
jismiya@qbs-generic-lin-2204:~/LINUX/pr_info$ sudo dmesg |tail
jismiya@qbs-generic-lin-2204:~/LINUX/pr_info$ |

```

```

jismiya@qbs-generic-lin-2204:~/LINUX/1$ cat param.c
#include <linux/module.h>
#include <linux/kernel.h>
#include <linux/moduleparam.h>

int num = 0;
char *name = "Guest";

//module_param(num, int, 0644);
module_param(name, charp, 0644);

static int param_init(void)
{
    printk(KERN_INFO "Module Loaded\n");
    //    printk(KERN_INFO "num = %d\n", num);
    printk(KERN_INFO "name = %s\n", name);
    return 0;
}

static void param_exit(void)
{
    printk(KERN_INFO "Module Removed\n");
}

module_init(param_init);
module_exit(param_exit);

MODULE_LICENSE("GPL");
MODULE_AUTHOR("Jismiya");
MODULE_DESCRIPTION("Module Parameter Example");
jismiya@qbs-generic-lin-2204:~/LINUX/1$ sudo insmod param.ko name="JISMIYA"
jismiya@qbs-generic-lin-2204:~/LINUX/1$ sudo dmesg |tail
[91780.302501] Module Loaded
[91780.302505] num = 0
[91780.302506] name = JISMIYA
jismiya@qbs-generic-lin-2204:~/LINUX/1$ sudo rmmod param
jismiya@qbs-generic-lin-2204:~/LINUX/1$ sudo insmod param.ko name="JISMIYA"
jismiya@qbs-generic-lin-2204:~/LINUX/1$ sudo dmesg |tail
[91780.302501] Module Loaded
[91780.302505] num = 0
[91780.302506] name = JISMIYA
[91787.185754] Module Removed
[91788.828468] Module Loaded
[91788.828482] num = 0
[91788.828483] name = JISMIYA
jismiya@qbs-generic-lin-2204:~/LINUX/1$ sudo rmmod param
jismiya@qbs-generic-lin-2204:~/LINUX/1$ sudo dmesg |tail
[91780.302501] Module Loaded
[91780.302505] num = 0
[91780.302506] name = JISMIYA
[91787.185754] Module Removed
[91788.828468] Module Loaded
[91788.828482] num = 0
[91788.828483] name = JISMIYA
[91796.155807] Module Removed
jismiya@qbs-generic-lin-2204:~/LINUX/1$ sudo dmesg -C
jismiya@qbs-generic-lin-2204:~/LINUX/1$ sudo dmesg -C
jismiya@qbs-generic-lin-2204:~/LINUX/1$ |

```

```

jismiya@qbs-generic-lin-2204:~/Linux/strace$ cat stack_trace.c
#include <linux/module.h>
#include <linux/kernel.h>

void function3(void)
{
    pr_info("Inside Function3\n");

    dump_stack();
}

void function2(void)
{
    pr_info("Inside Function2\n");

    function3();
}

void function1(void)
{
    pr_info("Inside Function1\n");

    function2();
}

static int __init my_init(void)
{
    pr_info("Module Loaded\n");

    function1();

    return 0;
}

void function4(void)
{
    dump_stack();
}

static void __exit my_exit(void)
{
    pr_info("Module Removed\n");
}

module_init(my_init);
module_exit(my_exit);

MODULE_LICENSE("GPL");
jismiya@qbs-generic-lin-2204:~/Linux/strace$ sudo insmod stack_trace.ko
jismiya@qbs-generic-lin-2204:~/Linux/strace$ sudo dmesg |tail
[112152.275290] entry_SYSCALL_64_after_hwframe+0x78/0xb8
[112152.275292] RIP: 0033:0x7767c531e90d
[112152.275298] Code: 5b 41 5c c3 66 0f 1f 84 00 00 00 00 00 f3 0f 1e fa 48 89 f8 48 89 f7 48
[112152.275299] RSP: 002b:00007ffdf38906f88 EFLAGS: 00000246 ORIG_RAX: 0000000000000139
[112152.275301] RAX: ffffffffdfda RBX: 000055c84cc41780 RCX: 00007767c531e90d
[112152.275302] RDX: 0000000000000000 RSI: 000055c8225e4cd2 RDI: 0000000000000003
[112152.275303] RBP: 0000000000000000 R08: 0000000000000000 R09: 0000000000000000
[112152.275303] R10: 0000000000000003 R11: 0000000000000246 R12: 000055c8225e4cd2
[112152.275304] R13: 000055c84cc45770 R14: 000055c8225e3888 R15: 000055c84cc41890
[112152.275305] </TASK>
jismiya@qbs-generic-lin-2204:~/Linux/strace$ sudo rmmod stack_trace
jismiya@qbs-generic-lin-2204:~/Linux/strace$ sudo dmesg |tail
[112152.275292] RIP: 0033:0x7767c531e90d
[112152.275298] Code: 5b 41 5c c3 66 0f 1f 84 00 00 00 00 00 f3 0f 1e fa 48 89 f8 48 89 f7 48
[112152.275299] RSP: 002b:00007ffdf38906f88 EFLAGS: 00000246 ORIG_RAX: 0000000000000139
[112152.275301] RAX: ffffffffdfda RBX: 000055c84cc41780 RCX: 00007767c531e90d
[112152.275302] RDX: 0000000000000000 RSI: 000055c8225e4cd2 RDI: 0000000000000003
[112152.275303] RBP: 0000000000000000 R08: 0000000000000000 R09: 0000000000000000
[112152.275303] R10: 0000000000000003 R11: 0000000000000246 R12: 000055c8225e4cd2
[112152.275304] R13: 000055c84cc45770 R14: 000055c8225e3888 R15: 000055c84cc41890
[112152.275305] </TASK>
[112187.668960] Module Removed
jismiya@qbs-generic-lin-2204:~/Linux/strace$ sudo dmesg -C
jismiya@qbs-generic-lin-2204:~/Linux/strace$ sudo dmesg -C

```

```

jismiya@qbs-generic-lin-2204:~/Linux/Thread$ cat kthread_demo.c
#include <linux/module.h>
#include <linux/kernel.h>
#include <linux/kthread.h>
#include <linux/delay.h>

static struct task_struct *thread1;

int thread_function(void *data)
{
    while (!kthread_should_stop())
    {
        pr_info("Kernel Thread Running\n");

        ssleep(2);
    }

    pr_info("Kernel Thread Stopped\n");

    return 0;
}

static int __init my_init(void)
{
    pr_info("Module Loaded\n");

    thread1 = kthread_run(thread_function,
        NULL,
        "my_kthread");

    return 0;
}

static void __exit my_exit(void)
{
    kthread_stop(thread1);

    pr_info("Module Removed\n");
}

module_init(my_init);
module_exit(my_exit);

MODULE_LICENSE("GPL");
jismiya@qbs-generic-lin-2204:~/Linux/Thread$ sudo insmod kthread_demo.ko
jismiya@qbs-generic-lin-2204:~/Linux/Thread$ sudo dmesg |tail
[113712.279610] Module Loaded
[113712.279675] Kernel Thread Running
[113714.299516] Kernel Thread Running
[113716.347705] Kernel Thread Running
[113718.395496] Kernel Thread Running
[113720.443715] Kernel Thread Running
[113722.491549] Kernel Thread Running
jismiya@qbs-generic-lin-2204:~/Linux/Thread$ sudo rmmod kthread_demo
jismiya@qbs-generic-lin-2204:~/Linux/Thread$ sudo dmesg |tail
[113718.395496] Kernel Thread Running
[113720.443715] Kernel Thread Running
[113722.491549] Kernel Thread Running
[113724.539594] Kernel Thread Running
[113726.587625] Kernel Thread Running
[113728.635606] Kernel Thread Running
[113730.683650] Kernel Thread Running
[113732.731706] Kernel Thread Running
[113734.779698] Kernel Thread Stopped
[113734.779757] Module Removed
jismiya@qbs-generic-lin-2204:~/Linux/Thread$ sudo dmesg -C
jismiya@qbs-generic-lin-2204:~/Linux/Thread$ sudo dmesg
jismiya@qbs-generic-lin-2204:~/Linux/Thread$ |

```

```
jismiya@qbs-generic-lin-2204:~/LINUX/UTS$ cat uts.c
#include <linux/module.h>
#include <linux/kernel.h>
#include <generated/utsrelease.h>

static int __init my_init(void)
{
    pr_info("Kernel Release = %s\n", UTS_RELEASE);
    return 0;
}

static void __exit my_exit(void)
{
    pr_info("Module Removed\n");
}

module_init(my_init);
module_exit(my_exit);

MODULE_LICENSE("GPL");
jismiya@qbs-generic-lin-2204:~/LINUX/UTS$ sudo insmod uts.ko
jismiya@qbs-generic-lin-2204:~/LINUX/UTS$ sudo dmesg |tail
[112673.295968] Kernel Release = 6.8.0-124-generic
jismiya@qbs-generic-lin-2204:~/LINUX/UTS$ sudo rmmod uts
jismiya@qbs-generic-lin-2204:~/LINUX/UTS$ sudo dmesg |tail
[112673.295968] Kernel Release = 6.8.0-124-generic
[112700.016728] Module Removed
jismiya@qbs-generic-lin-2204:~/LINUX/UTS$ sudo dmesg -C
jismiya@qbs-generic-lin-2204:~/LINUX/UTS$ sudo dmesg
jismiya@qbs-generic-lin-2204:~/LINUX/UTS$ |
```

```

jismiya@qbs-generic-lin-2204:~/LINUX/Version_magic$ ls
hello.c hello.ko hello.mod hello.mod.c hello.mod.o hello.o Makefile modules.order Module.symvers
jismiya@qbs-generic-lin-2204:~/LINUX/Version_magic$ cat hello.c
#include <linux/module.h>
#include <linux/kernel.h>

static int __init my_init(void)
{
    printk(KERN_INFO "Version Magic Demo\n");
    return 0;
}

static void __exit my_exit(void)
{
    printk(KERN_INFO "Module Removed\n");
}

module_init(my_init);
module_exit(my_exit);

MODULE_LICENSE("GPL");

//modinfo hello.ko
//uname -r
//kernel check version if sam it will load or elso kernel format wrong error will print
jismiya@qbs-generic-lin-2204:~/LINUX/Version_magic$ sudo insmod hello.ko
jismiya@qbs-generic-lin-2204:~/LINUX/Version_magic$ modinfo hello.ko
filename:          /home/jismiya/LINUX/Version_magic/hello.ko
license:           GPL
srcversion:        5314AD5A2C0F85BF3CFA782
depends:
retpoline:         Y
name:              hello
vermagic:          6.8.0-124-generic SMP preempt mod_unload modversions
jismiya@qbs-generic-lin-2204:~/LINUX/Version_magic$ sudo dmesg |tail
[91150.684435] Version Magic Demo
jismiya@qbs-generic-lin-2204:~/LINUX/Version_magic$ sudo rmmod hello
jismiya@qbs-generic-lin-2204:~/LINUX/Version_magic$ sudo dmesg |tail
[91150.684435] Version Magic Demo
[91176.946100] Module Removed
jismiya@qbs-generic-lin-2204:~/LINUX/Version_magic$ sudo dmesg -c
[91150.684435] Version Magic Demo
[91176.946100] Module Removed
jismiya@qbs-generic-lin-2204:~/LINUX/Version_magic$ sudo dmesg -c
jismiya@qbs-generic-lin-2204:~/LINUX/Version_magic$ |

```

```

jismiya@qbs-generic-lin-2204:~/LINUX/WARN_ON$ sudo insmod warn.ko
insmod: ERROR: could not insert module warn.ko: File exists
jismiya@qbs-generic-lin-2204:~/LINUX/WARN_ON$ ls
Makefile modules.order Module.symvers warn.c warn.ko warn.mod warn.mod.c warn.mod.o warn.o
jismiya@qbs-generic-lin-2204:~/LINUX/WARN_ON$ sudo rmmod warn
jismiya@qbs-generic-lin-2204:~/LINUX/WARN_ON$ ls
Makefile modules.order Module.symvers warn.c warn.ko warn.mod warn.mod.c warn.mod.o warn.o
jismiya@qbs-generic-lin-2204:~/LINUX/WARN_ON$
jismiya@qbs-generic-lin-2204:~/LINUX/WARN_ON$ ls
Makefile modules.order Module.symvers warn.c warn.ko warn.mod warn.mod.c warn.mod.o warn.o
jismiya@qbs-generic-lin-2204:~/LINUX/WARN_ON$ cat warn.c
#include <linux/module.h>
#include <linux/kernel.h>

static int __init my_init(void)
{
    int value = 10;

    pr_info("Before WARN_ON\n");

    WARN_ON(value == 10);

    pr_info("After WARN_ON\n");

    return 0;
}

static void __exit my_exit(void)
{
    pr_info("Module Removed\n");
}

module_init(my_init);
module_exit(my_exit);

MODULE_LICENSE("GPL");
jismiya@qbs-generic-lin-2204:~/LINUX/WARN_ON$ sudo insmod warn.ko
jismiya@qbs-generic-lin-2204:~/LINUX/WARN_ON$ sudo dmesg |tail
[112518.399612] Code: 5b 41 5c c3 66 0f 1f 84 00 00 00 00 00 f3 0f 1e fa 48 89 f8 48 89 f7 48 89 d6 48
[112518.399613] RSP: 002b:00007ffc9d538408 EFLAGS: 00000246 ORIG_RAX: 0000000000000139
[112518.399615] RAX: ffffffffefda RBX: 00005f1fba88f780 RCX: 00007c2cb2d1e90d
[112518.399615] RDX: 0000000000000000 RSI: 00005f1f9d47bcd2 RDI: 0000000000000003
[112518.399616] RBP: 0000000000000000 R08: 0000000000000000 R09: 0000000000000000
[112518.399616] R10: 0000000000000003 R11: 0000000000000246 R12: 00005f1f9d47bcd2
[112518.399617] R13: 00005f1fba893770 R14: 00005f1f9d47a888 R15: 00005f1fba88f890
[112518.399618] </TASK>
[112518.399619] ---[ end trace 0000000000000000 ]---
[112518.399620] After WARN_ON
jismiya@qbs-generic-lin-2204:~/LINUX/WARN_ON$ sudo dmesg -C
jismiya@qbs-generic-lin-2204:~/LINUX/WARN_ON$ sudo dmesg -C
jismiya@qbs-generic-lin-2204:~/LINUX/WARN_ON$ sudo dmesg |tail
jismiya@qbs-generic-lin-2204:~/LINUX/WARN_ON$ sudo rmmod warn
jismiya@qbs-generic-lin-2204:~/LINUX/WARN_ON$ sudo dmesg |tail
[112556.365327] Module Removed
jismiya@qbs-generic-lin-2204:~/LINUX/WARN_ON$ sudo dmesg -C
jismiya@qbs-generic-lin-2204:~/LINUX/WARN_ON$ sudo dmesg -C
jismiya@qbs-generic-lin-2204:~/LINUX/WARN_ON$ sudo dmesg
jismiya@qbs-generic-lin-2204:~/LINUX/WARN_ON$ |

```